Unit 3 - Knowledge Representation

What is Knowledge Representation?

Knowledge Representation (KR) is the field of AI dedicated to representing information about the world in a form that computers can utilize to solve complex tasks. It involves encoding knowledge so that AI systems can reason about it.

Types of Knowledge

- Declarative Knowledge - Facts and statements (what things are)
- Procedural Knowledge - How to do something (procedures)
- Meta Knowledge - Knowledge about knowledge
- Heuristic Knowledge - Rules of thumb from experience
- Structural Knowledge - Relationships between concepts

Knowledge Representation Techniques

Logical Representation

Uses formal logic (propositional and predicate logic) to represent facts and rules. Allows automated reasoning and theorem proving.

Semantic Networks

Graph-based representation where nodes represent concepts and edges represent relationships between them.

Frames

Data structures for representing stereotyped situations. Each frame has slots (attributes) with values.

Production Rules

IF-THEN rules that represent knowledge as condition-action pairs. Used in expert systems.

Ontologies

Formal representation of knowledge as a set of concepts within a domain and relationships between them.

Inference Engine

The inference engine is the component of an expert system that applies logical rules to the knowledge base to deduce new facts. It uses forward chaining (data-driven) or backward chaining (goal-driven) reasoning.

Expert Systems

Expert systems are AI programs that emulate the decision-making ability of a human expert. They consist of a Knowledge Base (facts and rules) and an Inference Engine (reasoning mechanism). Examples include MYCIN for medical diagnosis and DENDRAL for chemical analysis.